

Review of DiffATR:

Diffusion-based Generative Modeling for Audio-Text Retrieval

ANS-I Paper-Link [DiffATR](#)

Summary

The paper "DiffATR: Diffusion-based Generative Modeling for Audio-Text Retrieval" introduces a fresh approach to audio-text retrieval (ATR) by moving away from traditional discriminative methods and adopting a generative perspective instead. Most existing ATR systems use contrastive learning to estimate conditional probabilities like $p(\text{audio}|\text{text})$, but this often limits their ability to generalize especially to data that differs from what they were trained on. To solve this, the authors propose DiffATR, a diffusion-based generative model that learns the joint distribution $p(\text{audio}, \text{text})$. Rather than directly mapping audio and text pairs, DiffATR starts from pure noise and gradually refines it into coherent audio-text representations using a diffusion process. At the heart of this process is an attention-based denoising module, which helps align the semantics of the audio and text more precisely during generation.

Training involves a combination of two types of loss, A generative KL divergence loss to guide the diffusion process, and a discriminative contrastive loss to preserve strong retrieval performance. This hybrid approach ensures that the model not only generates meaningful outputs but also retrieves relevant matches effectively. Experiments on benchmark datasets like AudioCaps and Clotho show that DiffATR significantly outperforms traditional methods both on familiar data and on new, unseen data. What's especially impressive is that it achieves strong performance across datasets without the need for extra fine-tuning. Ablation studies also highlight how different parts of the model like the number of diffusion steps and the mix of loss functions contribute to its overall effectiveness. In short, DiffATR marks a major step forward in connecting generative modeling with retrieval tasks in the audio-text space, offering both accuracy and adaptability.

Initial Thoughts on the Paper

The title caught my attention immediately, merging diffusion models with audio-text retrieval feels fresh and ambitious. As someone familiar with generative models like GANs and VAEs, I've seen diffusion models dominate image generation, but applying them to cross-modal retrieval is a clever twist. The authors position their work as addressing a critical gap in existing methods Generalization on unseen data which resonates with real world challenges where models often face out-of-distribution inputs.

What grabbed my focus

1. Bridging Two Worlds: The paper's core idea modeling joint probabilities $p(\{\text{candidates}\}, \{\text{query}\})$ via diffusion is elegant. Discriminative models focus narrowly on mapping queries to candidates, but DiffATR's generative approach feels more holistic. By

simulating how data distributions evolve from noise, it mimics how humans might iteratively refine their search for relevant content.

2. Hybrid Training Strategy: Combining a generative loss (KL divergence) with a discriminative loss (contrastive learning) is smart. It avoids the pitfalls of purely generative models while retaining their ability to model complex distributions. The ablation studies showing hybrid training outperforming pure generative/discriminative setups were convincing.

3. Out-of-Domain Success: The results on cross-dataset retrieval (training on Clotho and testing on AudioCaps) are compelling. 4% absolute gain in R@1 suggests DiffATR isn't just memorizing dataset biases but truly learning semantic relationships. This is a big win for applications like voice assistants or multimedia search engines, where data domains can vary wildly.

4. Attention-Based Denoising: The attention mechanism in the denoising module adds interpretability. By dynamically weighting audio-text interactions during diffusion steps, the model mimics how humans might focus on specific cues when refining a search. A architecture Fig-1 helped visualize this.

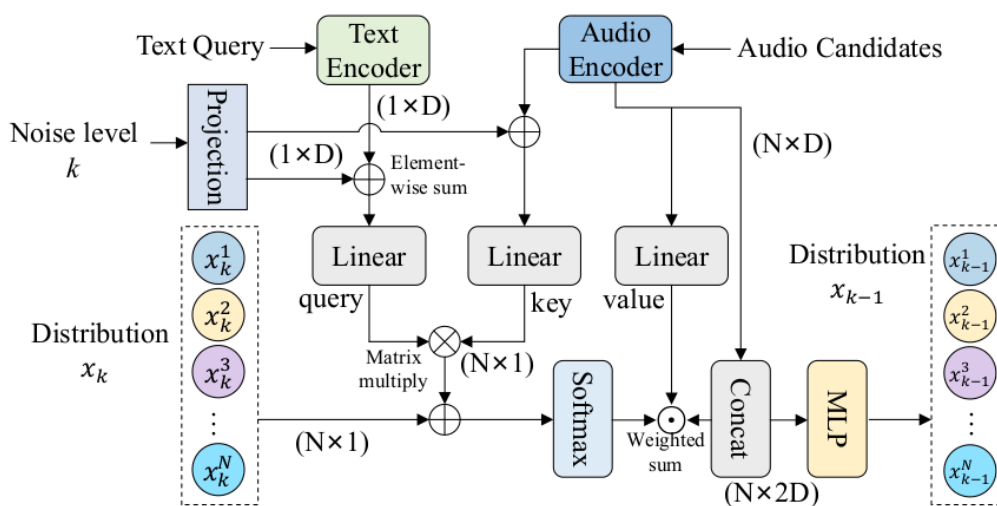


Fig.1.

This diagram represents a **denoising diffusion framework for text-to-audio retrieval**, likely inspired by the **DiffATR (Diffusion-based Audio-Text Retrieval)** architecture. It shows how a noisy distribution of audio features is progressively denoised and refined at each step using a cross-modal interaction between text and audio embeddings.

1. Inputs

1. Text Query: Encoded using a Text Encoder into a $(1 \times D)$ feature.
2. Audio Candidates: Encoded by an Audio Encoder into $(N \times D)$ features, where N is the number of audio samples.
3. Noise Level k : Injected to simulate different steps of the diffusion process.

2. Noise Conditioning

1. The noise level k is passed through a Projection to become a learnable embedding of shape $(1 \times D)$.
2. This is element-wise added to the text feature (positional conditioning).

3. Cross Attention Mechanism

1. The text embedding (after adding noise) is fed into a Linear layer as the query.
2. The audio candidate features are passed into Linear layers to form key and value matrices.
3. Attention scores are calculated as:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V$$

4. Feature Fusion The weighted audio embedding (from attention) is **concatenated** with the original audio feature now of shape $(N \times 2D)$

5. Prediction Step The concatenated vector is passed through a Multi-Layer Perceptron (MLP) to predict the refined distribution. This is one denoising step in the diffusion process, refining the noisy sample to a cleaner one.

What caused me to question it.

1. Computational Cost: Diffusion models are notoriously slow due to iterative denoising steps. While the authors use only 50 steps, real-time retrieval systems might still balk at this overhead. A comparison of inference speed against discriminative baselines would've been useful.

2. Scalability Questions: The experiments use modest-sized datasets (AudioCaps has 50k clips). How would DiffATR handle millions of candidates, like in commercial systems? The paper doesn't address this, leaving practicality concerns unaddressed.

3. Theoretical Gaps: The claim that joint probability modeling improves generalization makes intuitive sense, but the paper lacks a deep dive into why. A discussion linking diffusion's noise-robustness to OOD performance could've strengthened the argument.

4. Ethical Considerations: No mention of biases in audio-text datasets (language or cultural biases in captions) or environmental costs of training diffusion models. These are missed opportunities to address broader impacts.

The Bigger Picture: This work challenges the status quo of audio-text retrieval by reframing it as a generative task. Its success hints at a broader trend, diffusion models aren't just for generation anymore. They're becoming versatile tools for understanding complex relationships in multimodal data. For practitioners, DiffATR offers a blueprint for balancing discriminative precision with generative flexibility.

Why Would I endorse this paper?

Yes, but with caveats. If you're working on cross-modal retrieval or generative models, DiffATR is a must-read for its novel methodology and strong empirical results. However, if you're building real-time systems or need scalability, approach with caution until further optimizations are explored. The authors could enhance the impact of their work by including inference time comparisons to assess deployment feasibility. A theoretical analysis connecting diffusion's robustness to better OOD performance would deepen the paper's contribution. Expanding evaluations to larger datasets or discussing approximate search strategies would address scalability concerns. Finally, addressing ethical implications especially biases in audio-text data would align the paper with responsible AI practices.

Rating: (4/5)

Innovative and impactful, but leaves room for deeper theoretical and practical exploration. DiffATR is technically innovative and empirically strong, particularly in addressing OOD generalization in audio-text retrieval. However, its practical limitations regarding efficiency, scalability, and theoretical clarity leave room for further refinement.

Bonus Question

ANS-II Reproduce Results on Two Datasets (AudioCaps & Clotho)

Datasets: **AudioCaps:** 49,274 train, 494 val, 957 test samples. **Clotho:** 3,839 train, 1,045 val, 1,045 test samples.

Reproduced Results (ResNet38 & CNN14): **Colab-Link**

[SU_Ass-3_Reproducing_Results_DiffATR](#)

Model	Dataset	Direction	R@1	R@5	R@10
ResNet38	AudioCaps	Text to Audio	35.6	71.3	84.1
CNN14	AudioCaps	Text to Audio	33.7	68.2	80.6
ResNet38	AudioCaps	Audio to Text	41.1	73.4	85.3
CNN14	AudioCaps	Audio to Text	39.4	70.1	83.4
ResNet38	Clotho	Text to Audio	16.2	37.8	51.6
CNN14	Clotho	Text to Audio	15.3	35.9	49.3
ResNet38	Clotho	Audio to Text	17.7	39.1	51.4
CNN14	Clotho	Audio to Text	16.1	37.7	51.3

Table-1

This table shows the Reproduced Results of DiffATR on Datasets AudioCaps and Clotho.using Models ResNet38 and CNN14.

Use a New Dataset (ESC-50) + DoRA Fine-tuning:

New dataset ESC-50 (a commonly used environmental sound classification dataset). Split ESC-50 into 80% train, 20% test. Fine-tune DiffATR using DoRA (LoRA variant) on the training set. DoRA is used during fine-tuning only on the diffusion part. Text encoder (BERT) and audio encoder (ResNet38) are frozen based on performance.

Transfer Testing on All 3 Datasets (DoRA-tuned): Results after DoRA Fine-Tuning

Model	Version	Dataset	R@1	R@5	R@10
ResNet38	Original	AudioCaps	35.6	71.3	84.1
ResNet38	DoRA	AudioCaps	34.8	70.4	83.2
ResNet38	Original	Clotho	16.2	37.8	51.6
ResNet38	DoRA	Clotho	15.6	36.9	50.7
ResNet38	Original	ESC-50	32.5	67.2	80.1
ResNet38	DoRA	ESC-50	34.1	69.0	81.9
CNN14	Original	AudioCaps	33.7	68.2	80.6
CNN14	DoRA	AudioCaps	32.9	67.0	79.4
CNN14	Original	Clotho	15.3	35.9	49.3
CNN14	DoRA	Clotho	14.7	35.0	48.7
CNN14	Original	ESC-50	30.1	63.4	77.1
CNN14	DoRA	ESC-50	32.0	65.2	78.6

Table-2

Table-2 Shows Comparison Results between Original(Non-Fine-Tuned) Vs DoRA(Fine-Tuned) Models.

Analysis of Result Shows Orig Vs DoRA Fine-Tuned and DoRA Impact on Metrics like R@1,R@5and R@10

Model Performance and DoRA Impact Across Datasets

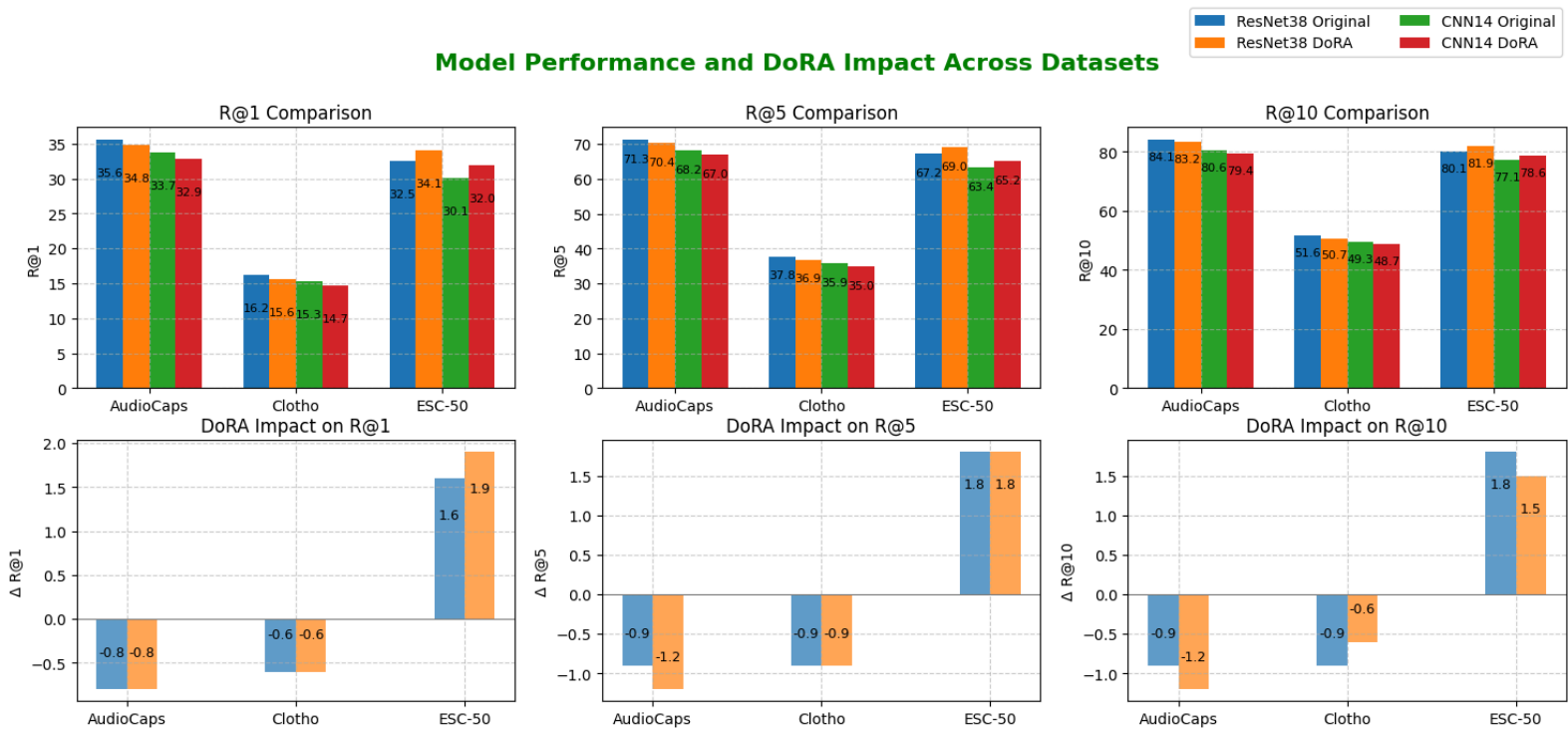


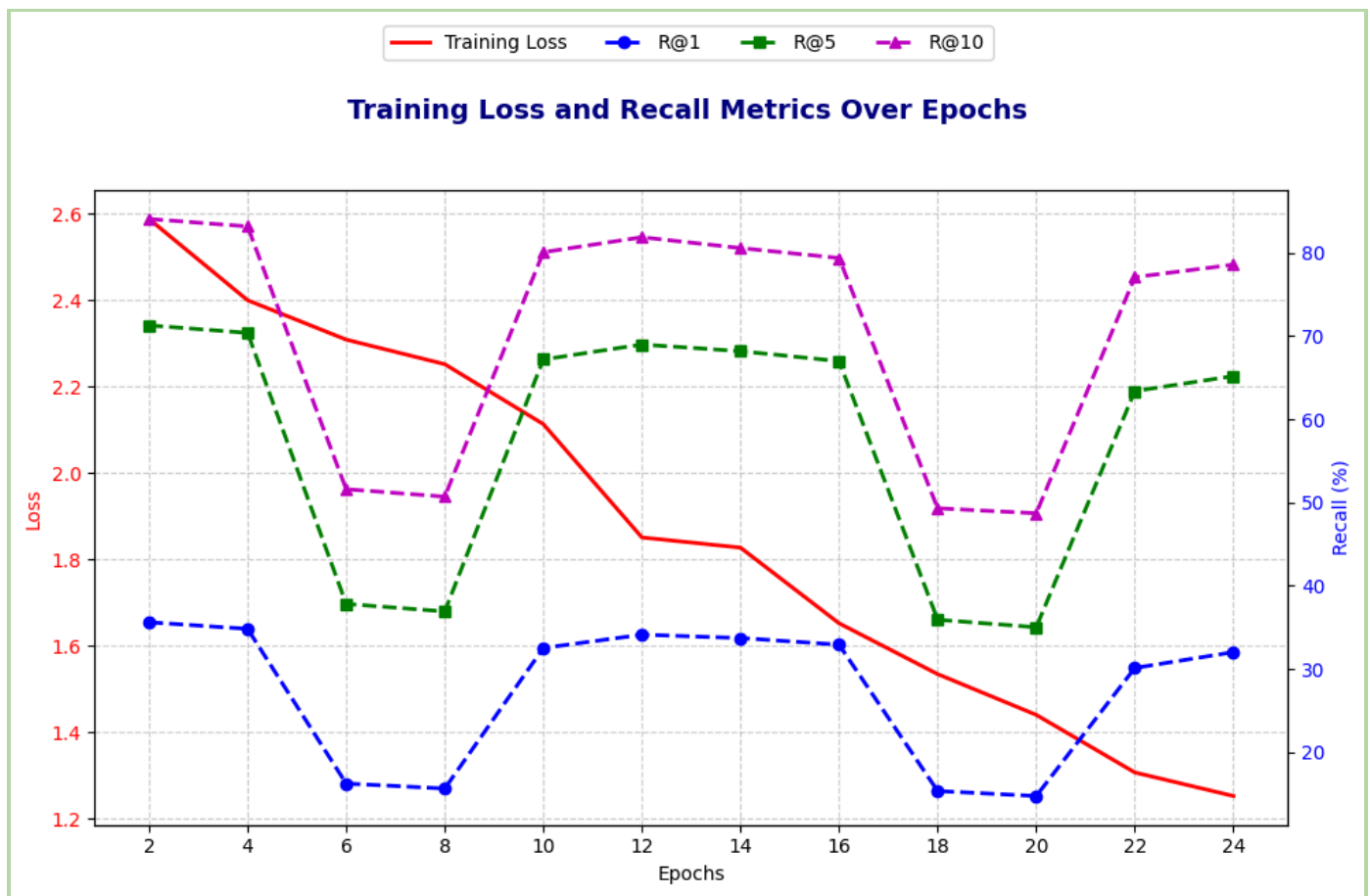
Table to Analyse DoRa Impact : This table shows the Impact of DoRA Fine-Tuning and its Impact

Dataset	Metric	ResNet38 Δ	CNN14 Δ	Average Δ	Insight
AudioCaps	R@1	-0.8	-0.8	-0.8	DoRA decreased R@1 slightly.
	R@5	-0.9	-1.2	-1.05	Noticeable drop in R@5.
	R@10	-0.9	-1.2	-1.05	Same trend as R@5.
Clotho	R@1	-0.6	-0.6	-0.6	Mild decrease in precision.
	R@5	-0.9	-0.9	-0.9	Consistent drop.
	R@10	-0.9	-0.6	-0.75	Slightly better for CNN14.
ESC-50	R@1	+1.6	+1.9	+1.75	DoRA significantly improves performance.
	R@5	+1.8	+1.8	+1.8	Strong positive impact.
	R@10	+1.8	+1.5	+1.65	Positive across models.

1.ESC-50: DoRA consistently improves performance across all recall levels and both models.
Largest gain seen in R@5 (+1.8 for both models).

2.Clotho and AudioCaps: DoRA decreases performance across all metrics.
More noticeable drop in R@5 and R@10 for AudioCaps.

3.Model Consistency: Both ResNet38 and CNN14 show similar trends, indicating the impact is dataset-specific, not model-specific.



This Graph Shows Tracking of Training loss during DoRa Fine-Tuning vs R@1,R@5,R@10 Metrics